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Entry in Monopoly Markets

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This paper develops new empirical models of market concentration from game-theoretic models of entry. We construct our models from inequality conditions that describe entrants' equilibrium strategies in simultaneous-move and sequential-move games, and use them to study the effects of entry in isolated monopoly markets for new automobiles. From estimates of the market size necessary to support one and two dealers, we conclude that monopoly dealers do not block the entry of a second dealer. We also find that entry does not cause price-cost margins to fall by much.

1. INTRODUCTION

Many technological and strategic forces shape market structure, including: economies of scale, cost differences among firms, entrants' expectations, first-mover advantages, and entry barriers. While previous empirical studies of entry have examined some of these determinants, these studies have not used the structure of game-theoretic models to model market structure. This paper develops new empirical models of market structure from game-theoretic models of entry. Our analysis focuses on the leading case of strategic entry—the entry of a second firm into a monopoly market.

We develop our empirical models of market structure from qualitative choice models of firms' entry decisions. These models presume that we do not observe entrants' revenues or costs. Instead, we draw inferences about firms' unobservable profits from threshold conditions that describe potential entrants' strategies. These threshold conditions resemble those used to construct single-person qualitative choice models (e.g. McFadden (1982)). Our models differ, however, because firms' profits depend on their competitors' actions. Section 2 develops a duopoly model that describes how potential entrants' unobserved *equilibrium* profits vary with: other entrants' strategies, the size of the market, and input costs. Section 3 relates these profit functions to simultaneous-move and sequential-move models of firms' entry decisions. From these theoretical models, we develop discrete dependent variable models of the number of firms that can "fit" in a market.

We use our econometric models to study market concentration in retail markets for new automobiles. From a model of dealer competition, we develop two measures of the effects of entry. One statistic summarizes the competitive cost of entry. The second statistic measures the presence of entry barriers or differences in entrants' fixed costs. We estimate these models using cross-section data on 149 geographically-isolated U.S. markets. These markets vary in size from a few hundred demanders and no dealers, to several thousand demanders and many dealers. By observing the market sizes at which dealers enter, we estimate the effect that entry has on dealer competition and monopoly profits. Both simultaneous-move and sequential-move models of this relationship suggest that the second entrant has nearly the same costs and market opportunities as the first entrant. In other words, we find no evidence that the second entrant faces entry barriers. We also find, however, that when the second dealer enters, the monopoly dealer's variable profits do not fall by much.

2. PROFIT IN MONOPOLY AND DUOPOLY

Our empirical analysis draws inferences about entrants' profits and market competition from qualitative information on potential entrants' decisions. To illustrate how we plan to draw inferences about firms' unobservable post-entry profits, consider what we know about potential entrants' profits when we observe a monopoly market. In a monopoly market, firm i enters because it expects to earn non-negative long-run profits, i.e. $\Pi_i^M \geqq 0$. The next most profitable potential entrant, firm j, does not enter because it expects to earn negative duopoly profits, i.e. $\Pi_j^D < 0$. In a duopoly market, firms i and j enter because they expect to earn non-negative duopoly profits, i.e. $\Pi_i^D \geqq 0$ *and* $\Pi_j^D \geqq 0$. Other potential entrants do not enter because they expect to earn negative triopoly profits. Continuing on, in markets with N firms, N firms expect non-negative profits and all other potential entrants expect negative profits. Following the logic of single-person qualitative choice models, we use these inequality conditions to model potential entrants' unobserved profits. Our models address two new modelling issues. First, in contrast to models of individual choice, entrants make interrelated decisions. Second, firms' *unobserved* profits vary in response to competitors' actions. Following McFadden's (1981) analysis of consumer choice models, this section develops models of firms' profit functions from a theory of profit maximization.

We begin by assuming that potential entrants have demand functions of the form $Q_i = D_i(Z, P) \times S(Y)$, or in vector notation,

$$Q = D(Z, P) \times S(Y). \tag{1}$$

The function $D_i(Z, P)$ is a representative consumer's demand for product i. The scalar variable S equals the number of representative consumers. The vector of per capita demands D depends on the vectors Z and P, which respectively describe market conditions and potential competitors' prices. We assume that Z does not include S. That is, we assume that the size of the market does not affect consumers' tastes. This latter assumption implies that holding prices constant, a doubling of S will double unit sales. We express the size of the market, S, as a function of a vector Y to suggest that demographic variables may affect firms' assessments of the "size" of their market.

On the cost side, we assume firms' costs have the form:

$$C_i(Q_i, W) = c_i(W)Q_i + F_i(W). \tag{2}$$

The variable Q_i represents the unit sales of firm i. This cost function presumes that firms have constant marginal costs of $c_i(W)$ and fixed costs of $F_i(W)$. The vector W represents exogenous variables affecting costs, such as input prices. The subscripts on cost allow firms to have different fixed or variable costs of production.

By inverting the demand functions in (1), we obtain the following expression for potential entrant i's profits as a function of its output and its competitors' outputs:

$$\Pi_i = [P_i(Z, Q/S) - c_i]Q_i - F_i. \tag{3}$$

In a market with two potential entrants, this profit function depends on the firm's own output and its competitor's output. If firm i's competitor, firm j, chooses not to enter, then we evaluate the vector Q/S at $Q_j = 0$. Because our empirical analysis models entry in response to variations in S, our analysis focuses on how changes in S affect firms' equilibrium profits. In other applications, one may wish to focus instead on the exogenous variables W and Z.

In a monopoly market, firm i maximizes its profits by setting $\partial\Pi_i/\partial Q_i = 0$. Because this first-order condition only depends on Q_i and S through the ratio Q_i/S, the monopolist's output increases in proportion with S. This implies that the monopolist's sales per customer, Q_i/S, do not change with S. Similarly, for a large class of duopoly models, the first-order conditions for profit maximization only depend on firms' outputs through the vector Q/S. Consequently, equilibrium prices and Q/S do not depend on S. Together these results imply that firms' ***equilibrium profits*** increase linearly in S. That is, equilibrium profits equal

$$\begin{aligned}\Pi_i^N &= [P_i^N(Z, W) - c_i(W)]D_i^N(Z, W) \times S(Y) - F_i(W) \\ &= V_i^N(Z, W)S(Y) - F_i(W), \end{aligned} \tag{4}$$

where Π^N denotes monopoly $(N = M)$ or duopoly $(N = D)$ profits. Notice that while profits increase linearly with S, the slope of profits in S changes with N.

To understand how we propose to use the comparative statics of equilibrium variable profits in S, consider the equilibrium profit functions depicted in Figure 1. Figure 1 plots the monopoly profit function for firm i, and the duopoly profit function for its competitor, firm j. In this figure, markets with insufficient demand to cover firm i's fixed costs occur in the region where $\Pi_i^M < 0$. At the breakeven market size S^M, firm i covers its fixed costs and enters (providing firm j has not yet entered). Firm i has a monopoly since firm j perceives that $\Pi_j^D < 0$ at S^M. Many factors affect firm j's perceptions, including the amount of competition it expects from firm i should it enter. While firm j's perceptions keep it from entering at market sizes near S^M, as the market grows firm j eventually finds it profitable to enter. In Figure 1, this occurs at S^D where $\Pi_j^D = 0$. The duopoly breakeven level of demand S^D lies to the right of S^M because firm j has higher fixed costs and less variable profit per customer. The greater these disadvantages, the further S^D lies to the right of S^M.

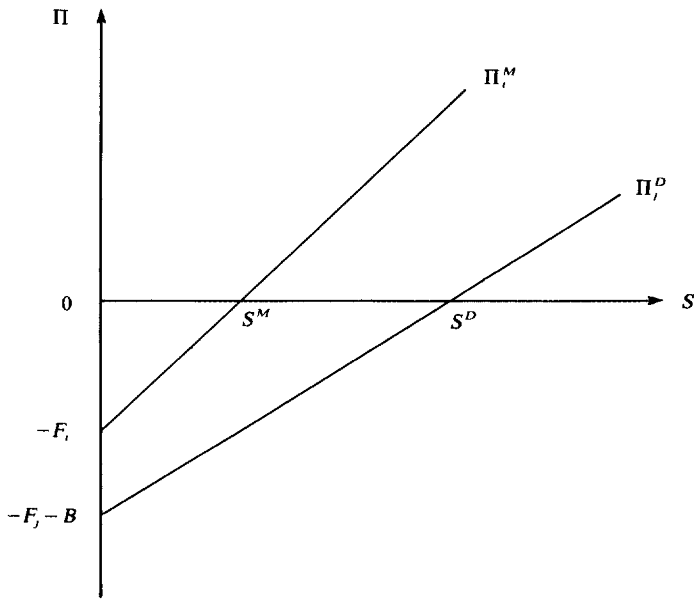

FIGURE 1
Profit as a function of market size

While economists typically do not observe firms' fixed costs or variable profits, they can estimate the breakeven levels of market demand, S^M and S^D. Suppose, for example, we observe a sample of markets that vary in S. By ranking the markets with respect to S, we can estimate the market size needed to support N firms. To interpret the monopoly and duopoly breakeven market sizes, we return to equation (4). At S^M, $\Pi_i^M(S^M)=0$. Because equilibrium profits increase linearly in S,

$$S^M = \frac{F_i}{(P_i^M - c_i)D_i^M}. \tag{5}$$

The duopoly breakeven market size S^D differs from S^M because of competition and differences in firms' products and costs. Suppose that firm j has higher costs and faces entry barriers as in Figure 1. Without loss of generality, we assume, as does Stigler (1968), that we can represent demand-reducing entry barriers as cost-increasing entry barriers. Suppose that these entry barriers raise j's marginal costs by b and its fixed costs by B. Firm j therefore has total fixed costs of $F_j + B$ and average variable costs of $c_j + b$. Setting $\Pi_j^D(S^D)=0$, we obtain

$$S^D = \frac{F_j + B}{(P_j^D - c_j - b)D_j^D}. \tag{6}$$

Comparing the monopoly breakeven level of demand in (5) to the duopoly breakeven level of demand in (6), we see that the ratio

$$\frac{S^M}{S^D} = \frac{(P_j^D - c_j - b)D_j^D}{(P_i^M - c_i)D_i^M}\,\frac{F_i}{F_j + B}$$

combines information about the *fraction by which variable profits fall* between monopoly and duopoly, and *the difference between the first and second entrant's fixed costs.*

The variable profit ratio

$$\frac{V^D}{V^M} = \frac{\partial \Pi_j^D/\partial S}{\partial \Pi_i^M/\partial S} = \frac{(P_j^D - c_j - b)D_j^D}{(P_i^M - c_i)D_i^M} \tag{7}$$

falls with increases in post-entry competition, unit production costs of the second firm, and product substitution. To calculate this ratio, we must know how firms' profits change with S. Although we do not observe profits, we can estimate the slope of profits in S from qualitative information on entrants' profits. Once we know V^D/V^M, we can then calculate the ratio of the entrants' fixed costs,

$$\frac{F^M}{F^D} = \frac{F_i}{F_j + B} \tag{8}$$

from an estimate of S^D/S^M.

Our analysis focuses on ratios of variable profits and fixed costs because our empirical models only identify firms' profits up to an unknown scale factor. We consider the specific ratios (7) and (8) because our data do not allow us to identify separately the individual components of these ratios. For example, a low value of (8) could suggest that the second entrant has higher fixed costs (i.e. $F_j > F_i$) or that it faces entry barriers (i.e. $B > 0$). Since we do not have information on entry barriers, we cannot separate F_j from B. From estimates of (8), however, we can ask whether cost differences among entrants exist; that is, we can test whether $F_i = F_j + B$.

Our discussion above contains several implicit assumptions about firms' costs and competition. For example, it assumes that entry barriers only affect the second entrant's profits. This assumption allows us to interpret S^M as a measure of the extent of scale economies. We do not include the cost of entry barriers in the monopolist's profit function Π^M because these costs do not affect the monopoly breakeven level of demand. A monopolist engages in entry deterrence only at long-run market sizes near S^D. Near S^M, the monopolist knows that it does not have to deter entry. Figure 2 illustrates this point graphically. At market sizes near S^D, the monopolist invests to raise potential entrants' costs. As drawn, these expenditures reduce the monopolist's profits and move S^D to the right. Although we do not observe the monopolist's expenditures on entry deterrence, S^M remains unchanged and S^D increases relative to that implied by a homogeneous entrant model. Consequently, entry deterrence by the monopolist lowers S^M/S^D.[1]

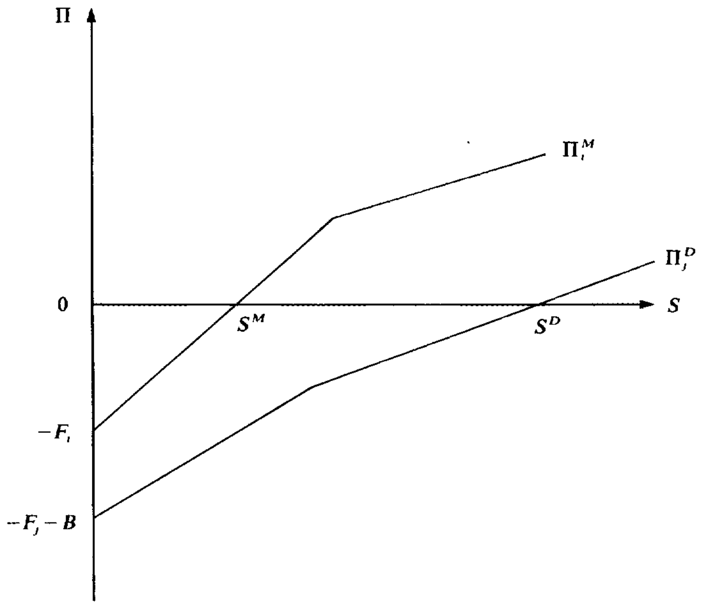

FIGURE 2
Profits with investments in entry deterrence

Finally, the duopoly model in this section contains many simplifying assumptions. It omits, for example, non-linear pricing strategies, asymmetric information, dynamic equilibrium concepts, etc. (See Milgrom and Roberts (1982), Roberts (1987), and Tirole (1988) for examples and extensions.) Each of these extensions would suggest new and richer interpretations of the firms' *equilibrium* profit functions. We do not explore these models here largely because our data do not permit us to distinguish among these alternative models. In principle, however, one could extend our analysis to consider more complicated games. Such extensions will likely change the relationship between market size and profits as described by equation (4), possibly making the relationship non-linear.

1. Monopolists sometimes recognize the costs of entry deterrence when making entry decisions. For example, Fudenberg and Tirole (1985) show that a monopolist may incur short-run losses to preempt a competitor. Firms, however, must recoup these short-run "investments" in the long-run if they are to break even.

3. EMPIRICAL MODELS OF STRATEGIC ENTRY

The previous section showed how firms' monopoly and duopoly profits depended on the size of the market, fixed costs, and entry barriers. This section uses firms' equilibrium payoff functions to formulate equations describing firms' optimal entry strategies. Following game-theoretic models of entry, we model entry as an all-or-nothing decision. Specifically, we allow entrant i one of two pure strategies: either enter, $I_i = 1$, or do not enter, $I_i = 0$. To simplify our econometric model, we model the decisions of just two potential entrants. These entrants play a one-shot entry game where each knows the strategies and payoffs of its competitor.

In this entry game, firm 1's entry decision depends on the action of firm 2. Similarly, firm 2's decision depends on the action of firm 1. To determine firms' equilibrium entry strategies, we must specify the timing of firms' decisions and the extent of firm cooperation. Here, we study the econometric implications of two widely used non-cooperative solution concepts, the simultaneous-move Nash and the sequential-move solution concepts. We concentrate on these two solution concepts because they illustrate most of the econometric issues that arise in entry games. (See also Bresnahan and Reiss (1990).)

3.1. *Simultaneous-move entry games*

The simultaneous-move Nash equilibrium solution concept defines a pure-strategy equilibrium as an action for each player such that no player can unilaterally increase their payoff. In our entry game, the pair of strategies I_1^* and I_2^* form a Nash pure-strategy equilibrium if

$$\Pi_1(I_1^*, I_2^*) \geqq \Pi_1(I_1, I_2^*)$$

for $I_1 \in \{0, 1\}$ and

$$\Pi_2(I_1^*, I_2^*) \geqq \Pi_2(I_1^*, I_2) \tag{9}$$

for $I_2 \in \{0, 1\}$. If firms earn zero profits when they play $I = 0$, then the following inequality system describes the players' optimal Nash entry strategies:

$$\begin{aligned} I_1^* = 0 \quad &\Leftrightarrow \quad (1 - I_2^*) \times \Pi_1^M + I_2^* \times \Pi_1^D < 0 \\ I_2^* = 0 \quad &\Leftrightarrow \quad (1 - I_1^*) \times \Pi_2^M + I_1^* \times \Pi_2^D < 0. \end{aligned} \tag{10}$$

For arbitrary monopoly and duopoly profits, these inequalities have several possible pure-strategy solutions. (We rule out mixed-strategy outcomes by assumption.) These inequalities also do not have a pure-strategy solution when $\Pi_i^M > 0 > \Pi_i^D$ for player i and $\Pi_j^D > 0 > \Pi_j^M$ for player j. Because duopoly profits never exceed monopoly profits in entry games, we rule out these payoffs in the models that follow.

Table 1 lists the pure-strategy solutions of (10). The first line describes the range of profits in which no entry occurs. The second pair describes the range of profits in which

TABLE 1

Pure-strategy outcomes of the simultaneous-move game

$\Pi_1^M < 0$, $\Pi_2^M < 0$	No Entrants
$\Pi_1^D > 0$, $\Pi_2^D > 0$	Duopoly
$\Pi_1^M > 0 > \Pi_1^D$, $\Pi_2^M > 0 > \Pi_2^D$	Firm 1 or 2 Monopoly
$\Pi_1^M > 0$, $\Pi_2^D < 0$	Firm 1 Monopoly
$\Pi_1^D < 0$, $\Pi_2^M > 0$	Firm 2 Monopoly

both firms enter. The last three sets of inequalities describe monopoly markets. The last two definitions implicitly include the statements "and not any of the previous events". The third statement shows that the Nash solution concept does not uniquely determine the identity of the monopolist. In the language of games, this middle region contains non-unique pure-strategy outcomes; either firm could enter in this region and establish a "natural monopoly" position. Such non-unique natural monopoly outcomes occur in many oligopoly models where firms make discrete decisions. Katz and Shapiro (1985), for example, describe such equilibria in a model where firms' technologies have increasing returns.

The presence of non-unique equilibria in game-theoretic models makes it impossible to use standard qualitative choice models to model entrants' profits. We could eliminate this region of non-unique outcomes by adding an equilibrium selection rule that ensures uniqueness. We also could assume the firms never have profits in this range. Alternatively, Heckman (1978), Schmidt (1981), and others have shown that one can uniquely solve linear discrete systems such as (10) if they have a recursive structure. If we made (10) recursive, however, monopoly profits would equal duopoly profits for at least one firm. As an alternative to these ad hoc procedures, we reinterpret our model as one that predicts $N = I_1 + I_2$, the number of entrants. By modelling N, instead of I_1 and I_2 individually, we treat the last three regions in Table 1 as one outcome.

3.2. *Sequential-move entry games*

The indeterminacies that arise in the Nash simultaneous-move entry model occur because players make their decisions simultaneously. In many markets firms enter sequentially. In sequential-move games, the first potential entrant recognizes that its decision affects the actions of later entrants. The first mover, for example, can preempt the second mover's entry into a natural monopoly market. The inequality conditions that describe the equilibria of a sequential-move game have a structure similar to those of the simultaneous-move game. When $\Pi^D \leqq \Pi^M$ and firm 2 moves first, these inequality conditions have the form:

$$\begin{aligned} I_1^* = 0 \quad &\Leftrightarrow \quad (1 - I_2^*) \times \Pi_1^M + I_2^* \times \Pi_1^D < 0 \\ I_2^* = 0 \quad &\Leftrightarrow \quad \begin{cases} \Pi_2^M < 0 \\ \text{or} \\ \Pi_2^D < 0 \leqq \Pi_2^M \text{ and } \Pi_1^D \geqq 0. \end{cases} \end{aligned} \tag{11}$$

To appreciate how these conditions differ from those of the simultaneous-move game, consider the payoff space depicted in Figure 3. This figure assumes that monopoly and duopoly payoffs differ by a positive constant Δ. The two solution concepts have the same duopoly and "no entry" payoff regions. They partition the monopoly outcomes differently however. The shaded centre region of the figure contains payoff pairs where either firm could enter as a monopolist in the simultaneous-move entry game. In the sequential-move game, firm 2 moves first and claims these monopolies. Thus, in contrast to the simultaneous-move model, the sequential-move model has unique pure-strategy outcomes.

3.3. *The treatment of unobservables*

The inequalities that define firms' simultaneous-move Nash and sequential-move equilibrium entry strategies both involve threshold conditions on entrants' monopoly and

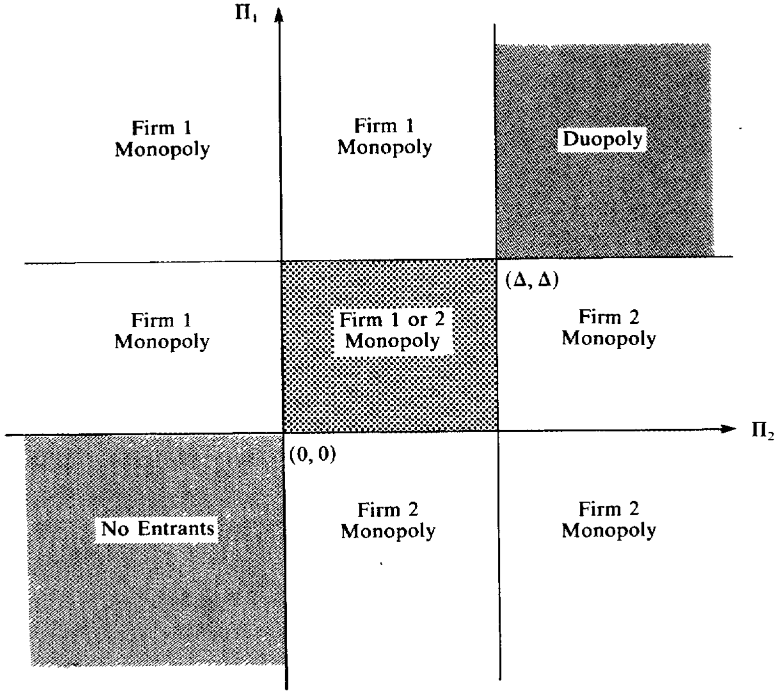

FIGURE 3

Model 2 entry outcomes $\Pi_i^D = \Pi_i^M - \Delta$

duopoly profits. Since we rarely observe firms' profits, much less firms' expectations about them, we model firms' profits as unobserved random variables. Following the structure of latent variables models, we include an additive error term in the equilibrium profit function (4). This error term represents that part of firms' profits that we cannot measure but that firms observe and act on. Specifically, we assume that the Nth entrant's profits have the form:

$$\begin{aligned}\Pi_i^N &= V_i^N \times S(Y) - F_i^N \\ &= [\bar{V}_i^N + \eta_i^N] \times S(Y) - \bar{F}_i^N - \varepsilon_i^N \\ &= \bar{\Pi}_i^N + \xi_i^N. \end{aligned} \tag{12}$$

This specification assumes that variable profits equal the sum of a measurable or expected variable profit $\bar{V}(\cdot)$ and an unobservable variable profit η. Our notation allows for the possibility that both measured and unobserved profits may change with the number of entrants. Fixed costs also have an additive form. The fixed cost error, ε, contains both unobserved differences in firms' fixed costs and entry barriers. As with variable profits, we assume that firms' fixed costs differ across markets and market structures.

In our econometric models, we restrict the stochastic structure of (12) for computational and economic reasons. If we did not restrict the distribution of ξ, then players' duopoly profits could exceed their monopoly profits with positive probability. To impose this constraint and yet still have computationally tractable models, we use several methods to constrain the variability of entrants' profits. Because our likelihood functions and model interpretations differ with each set of constraints, we now interpret each.

Model 1. *Perfectly correlated unobserved profits*

This model assumes that both potential entrants have the same unobservable fixed costs and variable profits—no matter how many firms enter. That is, this model assumes $\varepsilon_1^M = \varepsilon_1^D = \varepsilon_2^M = \varepsilon_2^D$ and $\eta_1^M = \eta_1^D = \eta_2^M = \eta_2^D$. While this error specification makes very strong assumptions about the distribution of unobserved profits, it has two primary advantages over more general specifications. First, by having only one error term for both firms, we obtain an ordered discrete dependent variable model for the equilibrium number of entrants, N. If, for example, firms only have independently and identically normally distributed fixed costs, then we can estimate firms' profit parameters using an ordered probit model. By also including an error term in variable profits, we obtain a heteroscedastic ordered probit model of N. In this second model, the variance of unobserved profits, $\sigma_\xi^2 = 1 + \sigma_\eta^2 S^2$, increases with market size. Besides their computational simplicity, these ordered probit models readily enable us to impose the constraint $\Pi_i^D \leqq \Pi_i^M$ at all market sizes.

To understand what effect the solution concept has on the estimation of these models, consider Figures 4(a) and 4(b). In these figures, we assume that the systematic components of profit, the $\bar{\Pi}_i^N$, differ across entrants. With only one error, ξ, the monopoly outcomes differ according to the ordering of firms' systematic profits. In both figures, no entry occurs for values of ξ to the left of min $[-\bar{\Pi}_1^M, -\bar{\Pi}_2^M]$. Similarly, both firms enter when ξ lies to the right of max $[-\bar{\Pi}_1^D, -\bar{\Pi}_2^D]$. Between these two bounds, we have three regions to consider. Between min $[-\bar{\Pi}_1^M, -\bar{\Pi}_2^M]$ and max $[-\bar{\Pi}_1^M, -\bar{\Pi}_2^M]$, the firm with the higher monopoly profit, firm 1, becomes the monopolist. Between max $[-\bar{\Pi}_1^M, -\bar{\Pi}_2^M]$ and min $[-\bar{\Pi}_1^D, -\bar{\Pi}_2^D]$, either firm could enter as a monopolist in the simultaneous-move game. In the sequential-move game the first mover preempts the second mover in this region. Finally, between min $[-\bar{\Pi}_1^D, -\bar{\Pi}_2^D]$ and max $[-\bar{\Pi}_1^D, -\bar{\Pi}_2^D]$ the firm with the greater duopoly profit maintains a monopoly.

Since in the simultaneous-move game we do not know which firm enters the natural monopoly region, we must treat the range of ξ between min $[-\bar{\Pi}_1^M, -\bar{\Pi}_2^M]$ and max $[-\bar{\Pi}_1^D, -\bar{\Pi}_2^D]$ as a single monopoly outcome. Further, because we do not know which firm earns the greatest profit in this range, we cannot separately identify firms' systematic profits. We therefore set $\bar{\Pi}_1^M = \bar{\Pi}_2^M$ and $\bar{\Pi}_1^D = \bar{\Pi}_2^D$. For a simultaneous-move

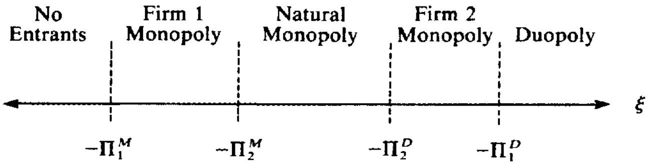

FIGURE 4(a)

Perfectly correlated error model outcomes

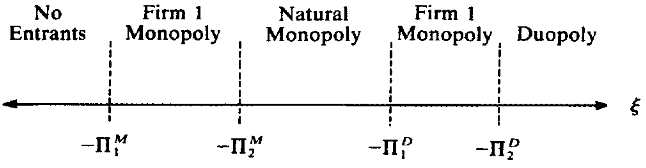

FIGURE 4(b)

Perfectly correlated error model outcomes model with normally distributed errors, we would calculate the probabilities of observing markets with no firms, two firms, or one firm as follows:

$$\begin{aligned} P_0 &= 1-\Phi[\bar{\Pi}^M(Z, W, Y)/\sigma_\xi] \\ P_2 &= \Phi[\bar{\Pi}^D(Z, W, Y)/\sigma_\xi] \\ P_1 &= 1-P_0-P_2, \end{aligned} \tag{13}$$

where $\Phi(\cdot)$ equals the cumulative distribution function of a standard normal random variable and we impose the constraint that $\bar{\Pi}^M \geqq \bar{\Pi}^D$. Probability statements for the sequential-move game have a form similar to (13), only we now assign the monopoly markets to first and second movers.

Model 2. *Correlated errors*

Model 2 generalizes Model 1 by allowing each firm to have its own unobservable profit component. To simplify the probability model for profits, we again assume that the unobserved component of profit does not change when market structure changes. That is, we assume that $\xi_i^M = \xi_i^D = \eta_i S - \varepsilon_i$. When each firm has a separate error term, we can decompose profits into two components: a market-specific and a firm-specific error component. That is, $\xi_i = \xi + \nu_i$. By assuming that each component is independent and identically distributed across markets, and identically distributed across firms, we can compute a measure of the importance of market-specific unobservables from the correlation coefficient $\rho = \text{var}(\xi)/(\text{var}(\xi)+\text{var}(\nu))$. Model 1, for example, assumes $\rho = 1$. When $\rho = 0$, entrants' profits do not have a market component.

In our empirical work, we estimate these correlated error models under the simplifying assumption that the firms' errors have a homoscedastic joint normal distribution. That is, we assume that entrants only have unobserved fixed costs. The probability of observing no entrants equals

$$P_0 = \Pr[\bar{\Pi}_1^M(Z, W, Y) < \varepsilon_1, \bar{\Pi}_2^M(Z, W, Y) < \varepsilon_2]. \tag{14}$$

A similar statement applies for P_2, with the inequalities reversed and duopoly profits replacing monopoly profits. The probability of observing a monopoly market in a simultaneous-move model equals the residual probability $P_1 = 1-P_0-P_2$. To compute these probabilities, we used the computational strategies discussed in Hausman and Wise (1978) and Daganzo (1979).

Model 3. *Monopoly and duopoly errors not perfectly dependent*

The previous two models assumed that each firm had the same unobserved profit term, whether or not the other firm entered. If a competitor's entry decision affects a firm's profits in unobserved ways, then we must alter the distributions of monopoly and duopoly profits. To obtain a simple model of the unobserved effects of entry, we again maintain that the dealers have the same unobserved profit term, ξ.[2] We now assume that unobserved duopoly profits equal $\varepsilon^D = \varepsilon^M + \varepsilon$ and $\eta^D = \eta^M - \eta$, where the random variables η and ε have supports on the positive real line. We restrict the supports of η and ε to insure that monopoly profits always exceed duopoly profits.

2. More general correlated error models raise few additional conceptual problems. These more general models, however, have much more complicated likelihood functions.

To estimate these models, we must find a simple distribution formula for the random components η^D and ε^D. If we assume that ε^M and η^M have independent normal errors and that ε and η have independent, positive half-normal distributions, then we can use results in Weinstein (1964) to compute the distributions of the sums ε^D and η^D. If each random component has a constant variance, then the probability of observing no entrants follows from (13). The probability of observing a duopoly equals

$$\begin{aligned} P_2 &= \Pr(\bar{\Pi}^D > \varepsilon^M + \varepsilon) \\ &= \int_{-\infty}^{\bar{\Pi}^D} \left[\int_0^{\bar{\Pi}^D - \varepsilon^M} pdf(\varepsilon) d\varepsilon \right] \phi(\varepsilon^M) d\varepsilon^M \\ &= -\Phi(\bar{\Pi}^D) + 2 \int_{-\infty}^{\bar{\Pi}^D} \Phi\left(\frac{\bar{\Pi}^D - \varepsilon^M}{\sigma_\varepsilon} \right) \phi(\varepsilon^M) d\varepsilon^M, \end{aligned} \tag{15}$$

where σ_ε^2 equals the variance of the half-normal error ε. The probability of observing a monopoly once again equals the residual probability $P_1 = 1 - P_0 - P_2$.

4. ENTRY IN AUTOMOBILE RETAILING MONOPOLY MARKETS

Having described an econometric framework for modelling entry and the equilibrium number of firms, we now use these models to study entry and competition among automobile dealers. Our interest in automobile retailers arose from our previous work on dealer markups (Bresnahan and Reiss (1985)) and recent federal studies of competition among dealers (e.g. Rogers (1986); see also the comments of Pashigian (1961), White (1971), and Smith (1982)). Most consumers in the United States buy their new cars from local car dealerships. Although these dealerships operate independently of the manufacturer, manufacturers spend considerable amounts recruiting and placing dealers. Both manufacturers and dealers claim that a dealer's success critically depends on the number of nearby dealers. With our entry models, we seek to find out what effect the entry of a second dealer has on competition.

To estimate our game-theoretic models, we assembled data on N, S, Y, Z, and W for a large sample of local markets. Because we could not clearly define market boundaries in urban areas, we instead collected data on dealers in isolated rural markets. We developed a list of candidate rural "markets" by canvassing all U.S. counties that had fewer than 10,000 people in 1980. In each county, we looked for an isolated town or centre of population that met the following two criteria:

(i) There was no other U.S. town with a population of over 1000 people within 25 miles of our central town.
(ii) There was no large U.S. city within 125 miles such that the city's population divided by the distance to our central town exceeded 600.

The first criterion eliminated over 100 candidate towns. For exmple, it eliminated Holdrege, Nebraska (5624 residents) and Minden, Nebraska (2939 residents), which lie roughly 20 miles apart. The second criterion reduces the possibility that our rural dealers would compete with a large urban dealership. For example, it excluded Pahrump, Nevada (100 residents), which lies 45 miles from Las Vegas, Nevada (164,674 residents). Besides these two criteria, we excluded any town that was part of an archipelago of towns. In four cases, we identified two towns within 10 miles of one another that were not part of an archipelago, and were at least 25 miles from other large towns. In these four cases, we defined the market as the two towns taken together.

Our search produced 149 markets or towns. These markets have anywhere from a few hundred to several thousand people. On average, the central towns in our markets lie 41 miles from the next town of more than 1000 people. Although our theory does not require that everyone in town must buy their car locally, our theory does assume that few consumers routinely shop outside town. Because residents of our central towns must often drive considerably more than 41 miles to reach the next town, we believe that the dealers in our sample towns have considerable market power.

We collected data on new car dealers in our markets from industry mailing lists, telephone directories, and site visits. We defined a *dealer* as a franchised new car dealer in operation at the end of 1982. Our sources indicated that 116 of our 149 markets had two or fewer dealers (see Table 2). We obtained information about towns in our sample from the *Rand McNally Commercial Atlas and Marketing Guide* and the 1980 *Census of Population and Housing*. We obtained the dealer data from R. L. Polk dealer mailing lists and Bell Telephone phone books. Polk's information comes from manufacturer franchise information. We checked Polk's lists against telephone directory records and visits to some of our towns. Our checks convinced us of the reliability of Polk's list.

In our 82 monopoly and duopoly markets, dealers sold either Ford Motor Company (F) or General Motors (G) cars. The Ford dealers always carried both Ford cars and trucks. The General Motors dealers usually carried Chevrolet cars and trucks. Some of our dealers also offered more expensive model lines, such as Mercury or Buick. Surprisingly, we found no "intercorporate" dual dealerships (i.e. dealers that carry more than one manufacturer's brand) among our monopolies and duopolies. In markets with three firms, the third dealer typically sold Chryslers. In our largest markets, thirteen dealers operated dual dealerships. All but one of these dual dealerships offered a domestic and a foreign car line.

Searches of Yellow Page advertisements and visits to some towns revealed that many of our dealers do more than just sell new cars. Some dealers, for example, sell new and used vehicles. Dealers also offer repair and towing services. Occasionally, dealers operate a non-automotive business, such as a farm equipment dealership. Since our dealers mainly sell cars, and opportunities for dealers to add sideline businesses do not appear to vary systematically across our markets, we believe that we have a reliable measure of the number of dealers.[3]

4.1. *Entry and market size*: *descriptive statistics*

The comparative statics in Section 2 showed how the breakeven market sizes S^M and S^D revealed information about firms' profits. We now estimate these breakeven levels of demand from data that describe how the number of entrants varies with market size. If central town population alone determines market size, then we can estimate S^M and S^D from cross-tabulations of town population by the number of dealers. Table 2 describes the results of such an exercise. It shows that towns with no dealer have an average 1980 population of 817 people; towns with a monopoly have an average population of 1514, and so on. The population percentiles in Table 2 show that 75% of the markets with no dealers have less than 924 residents; correspondingly, 75% of the markets with a monopoly dealer have at least 970 people. Thus, the breakeven level of demand for a monopolist

3. We also checked for multiple listings and multiple related businesses at the same address. For example, we matched Yellow Page information to Polk's dealer listings to distinguish between "Joe's Chevy" and "Joe's Used GM Cars". Obviously, we cannot know whether a single individual owns two adjacent businesses such as "Joe's Chevy" and "McGuire's Auto and International Harvester".

TABLE 2

Distribution of town population by the number of dealers in town

	No entrants	Monopoly	Duopoly	Three or more entrants
Number of markets	34	42	40	33
Sample percentile				
25th	488	970	1248	1814
30th	545	1052	1417	2016
35th	600	1070	1573	2115
Sample mean	817	1514	2282	2977
65th	824	1492	2579	3130
70th	896	1657	2949	3387
75th	924	1752	3154	3616

Notes. The town population data come from the *1980 Census of Population and Housing* and Rand McNally's *Commercial Atlas and Marketing Guide.* Dealer counts come from R. L. Polk's dealer mailing lists. See the text for sample definitions and counting rules.

probably lies somewhere between 925 and 975 people. Similarly, 65% of all monopolies have less than 1492 people, while 65% of all duopolies have more than 1573 residents. Thus, the breakeven level of demand for a duopoly probably lies somewhere between 1500 and 1600 people.

4.2. *Dealer profit functions*

Although the population percentiles in Table 2 show a strong, positive relationship between current town population and the number of dealers, other factors may affect dealers' assessments of market size. Dealers sell cars to people living outside the central town; they also take a long-run view of town population. These complications require a more complete model of $S(Y)$ and dealers' profits, $\Pi(S)$.

In our econometric models, we assume that S is a linear function of additional market size variables. Specifically, we assume that $S(Y) = Y\lambda$, where λ represents a column vector of known and unknown coefficients that transform Y into town population equivalents. We include 1980 town population, or *TOWNPOP*, in Y for obvious reasons. To capture dealer expectations about market growth, we include the asymmetric growth variables, *NGRW70* and *PGRW70.* These variables represent respectively the negative and positive growth in town population between 1970 and 1980. We include these two growth terms separately to account for the asymmetric way in which exit barriers or sunk entry costs may affect entry and exit decisions. Large sunk costs, for example, might keep firms from exiting when population unexpectedly declines. For a more complete discussion of the relation between expectations, irreversible investment, and forecasts of demand see Hause and du Rietz (1984).

Besides population growth variables, we include *OPOP10* in Y to measure the amount of demand from people living outside but near town. This variable is an estimate of the number of people living within ten miles of town. It includes residents of nearby towns and local residents not living within the borders of towns. We constructed our estimate of population not in town by assuming that non-town county residents were uniformly distributed throughout the county. In some markets, seasonal variations in population, Indian reservations, bodies of water, and other geographical peculiarities led us to modify our definition of *OPOP10.* We also experimented with several different definitions of *OPOP10*, including definitions that used an alternative distance criterion, such as people living within five, fifteen, twenty, and thirty miles of town. None of these other measures produced significantly different results and *OPOP10* usually had the greatest explanatory power. Finally, we included the variable *OCTY* in S to help us test our market selection criteria. The variable *OCTY* equals the number of county residents who regularly commute to jobs outside the county. It measures potential leakages in *TOWNPOP* because of low commuting costs.

To summarize our specification for $S(Y)$, we assume that dealers' expectations about the long-run size of the market have the form:

$$S(Y) = TOWNPOP + \lambda_1\, OPOP10 + \lambda_2\, NGRW70 + \lambda_3\, PGRW70 + \lambda_4\, OCTY. \quad (16)$$

We set the coefficient on *TOWNPOP* in (16) equal to one so that S has units equal to the number of people living in town in 1980. Finally, we chose not to include an error term in (16). By including a homoscedastic error term, we would make profits heteroscedastic in V. In practice, we found little evidence that V varied much across our markets. Thus, any homoscedastic error we included in S would have properties similar to ε.

Dealers' variable profits per customer, $V_i^N = (P_i^N - c_i^N)D_i^N$, could vary across markets for many reasons. We assume for simplicity that V^N varies linearly with a set of variables that describe local demand and cost conditions (i.e. W and Z). We also assume following equation (4) that *per capita* variable profits fall as N increases. Finally, because we do not differentiate among dealers in our simultaneous-move specifications, we assume that V^N does not differ across dealers. In sum, in simultaneous-move models we assume that per capita variable profits have the form

$$V^N = \theta^M + \theta^D D + Z\theta_Z + W\theta_W. \quad (17)$$

where the θ coefficients are unknown constants and D represents a zero-one dummy variable variable that equals one when the market has more than one firm. This specification presumes that duopoly profits differ from monopoly profits by the constant θ^D. We also estimated specifications in which the difference could vary across markets, but we found no evidence of intermarket differences.

We included a variety of variables in W and Z in an attempt to explain cross-section variations in local demand conditions and dealers' costs. These county-level variables included county residents': 1980 average income, median age, median years of schooling, and average retail wage. We also included: the fraction of residents who were female, the fraction of land in farms, the average value of farm land and buildings per acre, and the fraction of families that had a female head of household. Since many of these variables could affect both demand conditions and dealer costs, we do not draw inferences about demand and costs with these variables. We instead view the variable profit specification (17) as a reduced form model that summarizes intermarket variation in V^M and V^D.

Our model in Section 2 assumed that firms have constant marginal costs. We believe that to a first approximation our rural dealers have constant marginal costs. Several trade sources suggest that the largest component of dealer selling costs, the wholesale price of a car, does not change over most normal sales volumes. Davisson and Taggart (1974) also provide indirect evidence that other selling costs do not vary much over small sales volumes. While capacity constraints could affect dealer marginal costs, several industry studies suggest that dealers' unit costs do not increase over the small sales volumes of our dealers.

Although we have some evidence that suggests our dealers have constant marginal costs, our dealers clearly do not have the same unit costs. Wholesale car prices, the price of land, and the price of labour all vary across our markets. To proxy these cost differences, we collected data from the 1983 *County and City Data Book* on the county retail wage (*RETWAGE*) and the per acre value of agricultural land (*LANDVAL*). While these variables imperfectly measure factor prices, they are the best we could obtain. We include factor prices in variable profits to capture variation in unit costs. We include these prices in fixed costs to measure differences in dealers' fixed opportunity costs. Ideally we would like to include both an opportunity cost of time variable and a cost of capital variable in fixed costs. Since we do not have these variables, we assume that they vary randomly across our rural markets. Because later entrants may have higher fixed costs, we assume the second entrant's fixed costs increase by the constant γ^D. Specifically, we assume that fixed costs have the form

$$F^N = \gamma^M + \gamma^D D + \gamma_W RETWAGE + \gamma_L LANDVAL. \tag{18}$$

As in equation (17), because we do not observe profits, we cannot draw inferences about variable profits or costs from the estimated θ and γ coefficients. We can, however, draw inferences about profits with the ratios (6), (7), and (8).

5. EMPIRICAL RESULTS

Table 3 lists the variables we included in our specifications and their sources. Table 4 provides sample descriptive statistics. Table 5 presents estimates of Model 1, the simultaneous-move entry model where dealers have the same profit function. These ordered probit specifications assume that dealers only have unobservable fixed costs (i.e. $\eta_1 = \eta_2 = 0$). The first specification corresponds to the stylized profit functions depicted in Figure 1. It explains the number of firms using the 1980 population of the central town (*TOWNPOP*). Corresponding to the figure, we find the second entrant has higher fixed costs (i.e. $\gamma^D > 0$) and monopolists have greater variable profits (i.e $\theta^D < 0$).

The second specification in Table 5 adds variables to Y, the vector that describes the long-run size of the dealers' market. Population outside town increases demand. We

TABLE 3

Variable definitions

DTOTAL	Number of new car dealers in the market. (Polk)
TOWNPOP	Population of the central town in 1980. (CP and RM)
OPOP10	Population within 10 miles of the central town in 1980. (CP and RM)
NGRW70	Negative change in central town population between 1970 and 1980 (zero otherwise). (CP and RM)
PGRW70	Positive change in central town population between 1970 and 1980 (zero otherwise). (CP and RM)
OCTY	Number of people in the county who commute outside the county to work. (CCDB; series 89, 90, 108)
INCOME	*Per capita* money income in 1979. (CCDB; series 134)
RETWAGE	The 1977 average annual wage of paid employees in retailing in thousands of dollars. (CCDB; series 180 and 181)
DISTANCE	Distance from the central town to the next town with over 1000 people. (RM)
LANDVAL	Average market value of land and buildings per acre of farm land in 1978 in hundreds of dollars. (CCDB; series 209)
FARMFRAC	Fraction of land in agricultural uses. (CCDB; series 205)

Notes. Sources: R. L. Polk's dealer mailing list (1982), (Polk); the 1980 and 1970 *Census of Population and Housing*, (CP); the *Rand McNally Commercial Atlas and Marketing Guide*, (RM); and the computer tape edition of the 1983 *County and City Data Book*, (CCDB; the series numbers refer to series in the print edition).

TABLE 4

Sample descriptive statistics

Variable	Mean	Standard deviation
TOWNPOP	1885	1361
OPOP10	654	583
NGRW70	−58	116
PGRW70	168	353
OCTY	154	128
INCOME	5785	1113
LANDVAL	2·83	2·07
RETWAGE	5·22	0·93
DISTANCE	41	17
FARMFRAC	0·68	0·35
Sample Size	149	

Notes. In our empirical specifications we denominate the population and income variables in thousands.

TABLE 5

Ordered probit models of the number of dealers

Variable	Coefficient	Specification (1)	(2)	(3)	(4)
OPOP10	λ_1		0·254	0·196	0·208
			(1·26)	(0·99)	(1·15)
NGRW70	λ_2		2·666	2·682	2·692
			(3·45)	(3·40)	(3·41)
PGRW70	λ_3		−1·598	−1·421	−1·475
			(−4·94)	(−3·43)	(−4·07)
OCTY	λ_4		−0·645	−0·180	−0·188
			(−0·74)	(−0·22)	(−0·23)
V-Monopoly	θ^M	0·933	1·636	1·310	0·645
		(3·39)	(4·96)	(1·37)	(1·50)
V-Duopoly	$\theta^M + \theta^D$	0·702	0·965	0·673	−0·037
		(5·02)	(5·91)	(0·71)	(−0·12)
V-Income	θ_I			−0·008	
				(−0·09)	
V-RETWAGE	θ_W			−0·129	
				(−0·81)	
V-LANDVAL	θ_L			0·321	0·298
				(2·62)	(2·60)
V-FARMFRAC	θ_F			0·414	0·444
				(1·71)	(1·88)
F-Monopoly	γ^M	0·536	0·975	1·829	1·235
		(1·79)	(3·26)	(1·78)	(1·71)
F-Duopoly	$\gamma^M + \gamma^D$	1·277	1·434	2·384	1·749
		(5·39)	(5·11)	(2·24)	(2·42)
F-RETWAGE	γ_W			−0·323	−0·194
				(−1·64)	(−1·63)
F-LANDVAL	γ_L			0·367	0·341
				(2·01)	(1·96)
Log Likelihood		−123·76	−115·63	−108·44	−108·79

Note. Asymptotic *t*-statistics in parentheses.

estimate that it takes about four people living outside town to equal one person in town. Population changes over the past ten years affect long-run market size. Entrants respond to growth with a lag. Decreases in population significantly reduce predicted market size. We checked this latter prediction by comparing population changes between 1960 and 1970 with changes between 1970 and 1980 in our sample of towns. We found that while these population changes were positively correlated as our specifications suggest, they were not strongly correlated. We also included longer-run population changes in these specifications, but found that they did not change our estimates of $S(Y)$. Thus, we conclude that while population changes matter, our cross-section results cannot provide information on the timing of entry. Finally, the negative sign of the *OCTY* coefficient suggests that while commuters reduce demand, they do not do so by much.

The model in column (3) allows *per capita* demand and cost conditions to vary across localities. The specification in column (4) omits several insignificant variables from specification (3). Although the market-specific variables in these two specifications improve the model's fit, only the value of agricultural land has an individually significant effect at a 5% significance level. We also experimented with other demographic covariates, including: the median age of county residents; the proportion of county residents over 65; residents' median years of schooling; proportion of county residents who are female; and the proportion of county households headed by females. These variables explained little additional variation in profit, suggesting that our sample towns differ mainly in the number of demanders ($S(Y)$) and not demanders' tastes ($V(W, Z)$).

Besides the linear specifications reported in Table 5, we also estimated specifications that allowed market size to affect profits non-linearly. For example, we estimated models that included market size and squared market size. We also estimated specifications that allowed the effect of market size to vary with the minimum driving distance to the next town of over 1000 people (*DISTANCE*). These more general specifications do not reject the ordered probit models in Table 5. These models also imply similar breakeven market sizes.

We draw five main conclusions from the results in Table 5: (i) market size alone predicts much of the intermarket variation in the number of firms; (ii) the estimated relationship between market size and N does not change much when we add additional covariates; (iii) only a few demand and cost variables explain variation in N; (iv) market size depends on both current town population and market growth; and, (v) variable profits increase linearly with market size.

5.1. *Alternative stochastic specifications*

Table 6 presents more elaborate models of unobserved profits. As in Table 5, we base these models on the simultaneous-move entry game (10). Specification (5) of Table 6 reports estimates of a Model 1 ordered probit that has both fixed cost and variable profit error terms; that is, it sets $\xi = \varepsilon + \eta S$ with $\xi \sim N(0, 1 + \sigma_\eta^2 S^2)$. This extension does not produce dramatically different estimates of the λ, γ, or θ slope coefficients. At the means of the sample data, however, the unobserved variable profit error has a standard deviation of over 3 ($1{\cdot}688 \times 1{\cdot}8$), suggesting that most unobserved variation in profits comes from variable profits and not fixed costs.

Specification (6) relaxes the assumption that both entrants have the same unobserved profits. It assumes that the entrants have normally distributed fixed costs with unit variances and a bivariate correlation of $|\rho| < 1$ (Model 2). We estimate that ρ equals

TABLE 6

Extended models of the number of dealers

		Specification		
Variable	Coefficient	(5)	(6)	(7)
OPOP10	λ_1	0·126	0·208	0·143
		(0·70)	(1·15)	(0·82)
NGRW70	λ_2	1·344	2·688	2·658
		(1·83)	(3·39)	(3·53)
PGRW70	λ_3	−1·450	−1·473	−1·482
		(−4·08)	(−4·06)	(−4·36)
OCTY	λ_4	0·629	−0·197	−0·031
		(0·81)	(−0·24)	(−0·04)
V-Monopoly	θ^M	2·052	0·612	0·564
		(1·49)	(1·39)	(0·98)
V-Duopoly	$\theta^M + \theta^D$	0·157	−0·035	
		(0·16)	(−0·12)	
V-LANDVAL	θ_L	0·700	0·288	0·376
		(1·33)	(2·59)	(1·92)
V-FARMFRAC	θ_F	1·810	0·431	1·031
		(1·56)	(1·90)	(2·20)
F-Monopoly	γ^M	3·614	1·444	2·062
		(1·80)	(2·12)	(2·06)
F-Duopoly	$\gamma^M + \gamma^D$	4·443		
		(1·98)		
F-RETWAGE	γ_W	−0·411	−0·188	−0·298
		(1·53)	(−1·63)	(−1·94)
F-LANDVAL	γ_L	0·892	0·329	0·376
		(1·20)	(1·94)	(1·52)
	ρ		0·795	
			(3·06)	
	σ_η	1·688		
		(1·60)		
	σ_ε			1·664
				(2·86)
	σ_η			0·999
				(2·32)
Log Likelihood		−103·44	−108·78	−106·19

Note. Asymptotic *t*-statistics in parentheses.

0·795, which implies that unobservable firm-specific fixed costs account for only 20% of the unexplained variance in profits. Because of the large standard error on ρ and the small increase in the sample likelihood function, we conclude that most of the variation in profits occurs across markets and not among entrants within the same market. Thus, we do not reject the perfectly correlated error structure underlying specifications (1) through (5).

Specification (7) extends the model of unobserved profits to allow for unobserved shifts in firms' monopoly and duopoly profits (Model 3). This specification maintains perfect correlation among the dealers' errors, but allows monopoly and duopoly profits to differ by two positive half-normal random variables, one included in fixed costs and the other in variable profits. To simplify this simultaneous-move model, we attribute all differences in firms' monopoly and duopoly profit functions to random effects.[4] The large standard deviations of unobserved duopoly profits in Table 6 suggest that entrants'

4. When we estimated a model with γ^D and θ^D unconstrained, we found the derivative of the likelihood function with respect to either parameter was positive at zero. If we did not constrain these parameters, then the event $\Pi^D > \Pi^M$ would take on positive probability for some range of market sizes.

monopoly and duopoly profits differ substantially. The large estimated standard errors on the error variances and the log-likelihood value for this model, however, provide little reason to prefer this model of profit over the simpler specification in (4).

To summarize our assessments of the specifications in Table 6, none of these more complex treatments of unobserved profits improve on our simpler, homogeneous-entrant specification (4). We find evidence of unobserved variation in entrants' fixed costs and variable profits. We also find larger unobserved variation in variable profits. To develop a better understanding of why dealers' profit may differ, we checked the predictions of our models. Of the markets that had unusually high or low probabilities, three markets had "too few" dealers. Our biggest outlier was Munising, Michigan. In 1980, it had 3083 residents and yet no dealers. Phone calls to Munising and these other towns did not reveal obvious omitted economic variables. Instead, we heard of unusual local economic conditions (e.g. the closing of a local paper mill). As best we could tell, these anomalies were "market-wide" rather than entrant specific.

5.2. *Estimates of a sequential-move model*

Our simultaneous-move entry models in Tables 5 and 6 only allow dealers' profits to differ for unobservable reasons. We do observe, however, that dealers carry different model lines. For instance, 24 of our monopolists sell GM cars and 18 sell Ford cars. All duopolists sell either Ford or GM cars. The prevalence of monopoly GM dealerships may reflect consumers' preferences for GM cars or GM's superior dealer placement record. While we cannot separately distinguish between these two hypotheses with our data, we can estimate GM dealers' profit functions under the assumption that GM dealers have the first opportunity to enter a market. In this sequential-move game, GM would, all other things being equal, have more dealers because of its first-mover advantage in claiming natural monopoly markets.

We estimated a GM-leader model with a stochastic structure similar to that used in specification (6). In contrast to (6), we assumed that GM and Ford dealers profits differ in their fixed cost and variable profit intercepts. That is, we assumed that monopoly GM dealers' variable profits per customer equal

$$V_G^M = \theta_G^M + \theta_L \, LANDVAL + \theta_F \, FARMFRAC,$$

and Ford dealers' monopoly variable profits per customer equal

$$V_F^M = \theta_F^M + \theta_L \, LANDVAL + \theta_F \, FARMFRAC.$$

Table 7 presents estimates of this GM-leader model. The estimates reveal that GM dealers have lower fixed costs and Ford dealers have greater variable profits per customer. Because the estimated difference in fixed costs exceeds the difference in variable profits, GM dealers will enter smaller monopoly markets than Ford dealers. The small value of θ^D, however, implies that there are few "natural monopoly" markets where GM dealers preempt Ford dealers. The other parameter estimates do not differ much from those reported in specification (6). Our estimate of the correlation among dealers' profits decreases somewhat from specification (6); we now do not reject the null hypothesis that $\rho = 0$.

The log-likelihood value of the sequential-move model (8) in Table 7 is lower than the log-likelihood value of the simultaneous-move model (6) because model (6) does not

TABLE 7

Sequential-move models of the number of dealers

Variable	Coefficient	(8)
OPOP10	λ_1	0·228 (1·22)
NGRW70	λ_2	2·590 (2·10)
PGRW70	λ_3	−1·391 (−4·03)
OCTY	λ_4	−0·358 (−0·44)
V_G*-Monopoly*	θ_G^M	0·297 (0·78)
V_F*-Monopoly*	θ_F^M	0·543 (0·97)
V-LANDVAL	θ_L	0·281 (2·32)
F-FARMFRAC	θ_F	0·425 (1·91)
V-Duopoly	θ^D	0·402 (1·17)
F_G*-Monopoly*	γ_G^M	1·273 (1·76)
F_F*-Monopoly*	γ_F^M	1·554 (2·03)
F-RETWAGE	γ_W	−0·181 (−1·90)
F-LANDVAL	γ_L	0·281 (2·32)
	ρ	0·592 (0·658)
Log Likelihood		−136·52

Note. Asymptotic t-statistics in parentheses.

separate GM and Ford monopolies. To make the simultaneous-move and sequential-move log-likelihood functions comparable, we arbitrarily assigned natural monopoly markets to GM with a probability equal to the observed sample average of 24/42. This selection rule lowers the simultaneous-move log-likelihood values by 28·7—showing that the fits of the sequential-move and simultaneous-move models differ by small amounts.

5.3. *Monopoly and duopoly breakeven market sizes*

Table 8 reports the monopoly and duopoly breakeven market sizes implied by the specifications in Tables 5, 6, and 7. We calculated these breakeven ratios using equations (5) through (8), with the regressors evaluated at their sample means. Our calculations also correct for the non-zero means of the unobservables in Models 2 and 3. For Model 2, we condition the distribution of unobservable monopoly profits on the second-order statistic from the estimated bivariate normal distribution. For Model 3, we correct for the non-zero expected value of the asymmetrically distributed error. Although the specifications in Tables 5 and 6 have different parametric and stochastic structures, most imply a monopoly breakeven level of demand between 650 and 750 people. An estimated breakeven monopoly market size of 650 residents may not seem large enough for a dealer to survive. Aggregate data on dealer costs, prices, and demand, however, suggest that a

TABLE 8

Estimated breakeven monopoly and duopoly market sizes

Specification	Table	S^M	S^D	S^M/S^D	V^D/V^M	F^M/F^D
Model 1						
(1)	5	575	1820	0·316	0·752	0·419
		(188)	(166)	(0·095)	(0·180)	(0·183)
(2)	5	596	1486	0·401	0·590	0·680
		(133)	(178)	(0·070)	(0·139)	(0·194)
(3)	5	663	1518	0·437	0·642	0·680
		(135)	(171)	(0·070)	(0·143)	(0·184)
(4)	5	664	1538	0·432	0·619	0·698
		(135)	(173)	(0·068)	(0·135)	(0·184)
(5)	6	759	1433	0·530	0·640	0·828
		(103)	(141)	(0·057)	(0·111)	(0·163)
Model 2						
(6)	6	662	1538	0·431	0·624	0·691
		(166)	(175)	(0·069)	(0·141)	(0·192)
Model 3						
(7)	6	674	1893	0·356	0·656	0·542
		(115)	(251)	(0·067)	(0·106)	(0·122)
Sequential move						
(8)	7					
	GM	608	1595	0·381	0·709	0·538
		(180)	(220)	(0·170)	(0·158)	(0·196)
	FORD	671	1480	0·453	0·753	0·602
		(175)	(184)	(0·077)	(0·143)	(0·151)

Note. Asymptotic standard errors in parentheses. We evaluated the breakeven market sizes and standard errors at the means of the sample data. The breakeven market sizes have units equal to the number central town residents in 1980.

dealer could survive at this range of market sizes.[5] The estimates of the duopoly breakeven level of demand vary more, yet they do not differ much from the estimates obtained from Table 2.

The estimated ratios of breakeven market sizes vary slightly across specifications. Apart from specification (1), the ratios range between 0·4 and 0·53, suggesting that it takes about 2 to 2·5 times as much demand to support a second entrant as it does one. To interpret these ratios, we decomposed them into the ratio of variable profits, equation (7), and the ratio of fixed costs, equation (8). The ratio V^D/V^M measures the fraction by which variable profits per customer fall with the entry of the second firm. Evaluating this ratio at the sample means of the variables, we find that this ratio ranges from 0·62 to 0·66 in specifications (3) through (7). When duopolists sell the same product, V^D/V^M should be equal to or less than 1/2. For example, the ratio equals 1/2 when the duopolists collude and evenly divide profits. If the firms did not collude, the ratio would fall below 0·5. If the two firms sell different products, the ratio of variable profits could exceed 1/2. For example, the ratio equals one for products that are independent in demand.

5. A town of 650 people has approximately 425 adults between the ages of 18 and 64. On average, each adult in the U.S. purchases 0·95 new cars or light trucks every 11 years. Thus, a breakeven monopoly dealer would sell three cars per month or 36 cars each year, a small volume by urban standards. If our rural dealers sell their cars at the manufacturer's list price, they would earn about $2000 per vehicle, resulting in a $72,000 gross profit per year (see Bresnahan and Reiss (1985) for evidence on margins). This revenue must cover the cost of the dealer's building and lot, the dealer's time, and overhead expenses. If service departments operate as breakeven enterprises, then the dealer keeps about 1/3 of the $72,00, or $24,000, according to Davisson and Taggart (1974). Recalling that this calculation is for a *breakeven* dealer in a rural market, these figures seem plausible.

Since we find V^D/V^M greater than 0·5, we conclude that product differentiation increases duopoly margins more than competition lowers them. Unfortunately, we do not have price-cost margin data or sales figures that could confirm this hypothesis. National sales data suggest that product differences do affect margins and dealer volumes. In 1982, for example, GM dealers selling Chevrolets and another GM brand had 50% higher sales volumes than Ford-Lincoln-Mercury dealers. Thus, it seems plausible that entry by a Ford dealer into a monopoly GM market would not lower a monopoly GM dealer's sales and variables profits by much.

Finally, our estimates of the differences in firms' fixed costs vary across specifications. Specifications (3), (4), and (6) imply that the second entrant's fixed costs are about one and one-half times greater than the monopolist's fixed costs. Specification (5), which includes an error in variable profits, implies a lower value. Specification (7) on the other hand implies a much higher ratio. While these estimates suggest that the second entrant has higher fixed costs, we cannot necessarily attribute these costs to entry barriers. The sequential-move estimates, for example, suggest that fixed costs differ by brand and not just by the order of entry. Thus, we have little evidence to suggest that the second entrant has much higher fixed costs than the first entrant because of entry barriers.

6. CONCLUSION

This paper empirically modelled the effects of entry in monopoly and duopoly markets. Our models used the structure of game-theoretic models to model interrelations among potential entrants' profits and decisions. Specifically, we constructed empirical models from both simultaneous-move and sequential-move models of entrants' decisions. These models extend qualitative choice models to situations in which agents make interrelated decisions. With our models, we studied how shifts in market demand changed the number of automobile dealers that would "fit" in a concentrated retail market. We found that monopoly dealers do not delay the entry of a second dealer, as the second dealer enters at roughly two to two and one-half times the size of the market needed to support one firm. We also found that the entry of a second dealer does not cause variable profits or margins to fall by much. Our random effects specifications suggest, however, that entry patterns differ across markets, and that in some markets entry may not occur unless the market is much greater than twice the size of the monopoly breakeven level of demand.

We conclude by emphasizing that our models only begin the study of entry in oligopolistic markets. We did not consider many interesting solution concepts and entry games. For example, we did not model the dealer's choice of which car line to sell or the complications that arise when more than two dealers can enter. In future work, we will consider these complications and the relevance of our models to other economic decision problems (see Bresnahan and Reiss (1990)).

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